Project Description

Project Introduction

This project aims to develop a community-level predictive model for homelessness by analyzing data spanning 2010–2023. Unlike previous studies that primarily focused on predicting Homelessness at the individual level, this project centers on identifying structural community-level factors, such as rent levels and economic conditions, that influence homelessness. Using data from approximately 400 Continuums of Care (CoC) funded by the U.S. Department of Housing and Urban Development (HUD), the project seeks to predict the number or rate of homelessness in each CoC, providing insights to guide policy decisions and resource allocation.

Big Impact

Homelessness is a complex societal challenge shaped by a range of factors. The project seeks to develop models that can assess the influence of various factors on homelessness and accurately forecast homelessness rates, offering valuable insights to help policymakers and community organizations allocate resources more efficiently and strengthen the effectiveness of assistance programs. Through the prediction of homelessness rates and trends, the project seeks to enable timely interventions that can reduce homelessness on a larger scale.

Objectives

This semester, our team's two main objectives were to:

1. Identify and fix errors and inconsistencies in the dataset created by the previous semester's team
2. Develop more accurate machine learning models than last semester's team

Data Cleaning

We identified and fixed errors with the following data features: total beds, median rent, median income, and all five target variables.

1. Total Beds

   We observed that approximately 42% of the data was missing in the "total beds" column from last semester's dataset. This issue stemmed from an error in summing the bed counts for Emergency Shelters (ES), Transitional Housing (TH), and Safe Havens (SH).

Specifically, when any of these individual columns had a missing value, it was treated as NA rather than zero, resulting in the total summation returning NaN. To capture total bed count, we used a conservative approach, imputing zero beds for any missing bed values to avoid excluding records or inflating capacity estimates. This method ensured data integrity while providing a complete dataset for analysis. This approach reflects the reality that individuals may use different shelter types based on availability and need. Imputing zeros minimizes the risk of overestimating resources, which is crucial for informed policy planning and resource allocation.

2. Rent + Income

   After calculating the mean, median, maximum, and minimum values for the rent and income columns in last semester's dataset, we found the figures to be unrealistically high, with the mean rent around $186K and the mean income approximately $10.8M. Upon investigation, we discovered that the correct procedure for obtaining accurate rent and income data involves matching ACS rent and income data at the census tract level to their corresponding CoC (Continuum of Care) regions, which consist of multiple census tracts. To accurately represent the entire CoC, a population-weighted average must be calculated. However, the previous dataset had simply summed the values across census tracts, which led to inflated results. We corrected this by re-downloading the relevant ACS data: income [B19013_001E], rent [B25064_001E], and population [B01003_001E], and used them to compute the proper population-weighted averages. This is the equation we used to calculate the population weighted average of rent or income within one CoC:

$$\text{Weighted Average} = (\Sigma\,(\text{Value}_i \times \text{Population}_i))\,/\,\Sigma\,\text{Population}_i$$

3. Target Variables

   After calculating the mean, median, maximum, and minimum values for the target variables in last semester's dataset, we found the maximum values to be very high. We found out that the columns included a row for "total number of homeless in the country" which is why the number was so high. To fix this, we had to download the PIT dataset and remove the total rows for each year in the dataset. Once removed in Excel, we then imported the sheet to a Python notebook to transpose the data into one csv file with the corrected values.

   When predicting the target variables, it's crucial to normalize them per 1,000 people. Without this step, models tend to learn a trivial pattern — that areas with larger populations have more homeless individuals — which is not a meaningful insight. The previous team attempted normalization, but according to their final report, they did so by simply dividing each value by 1,000. The correct approach is to first divide the raw homelessness counts by the total population of the CoC, then multiply the result by 1,000. We applied this method, and this is the equation we used to normalize the target variables for each CoC:

$$\textit{Normalized Target} = (\textit{Raw Count} / \textit{Population}) \times 1{,}000$$

Model Building

Metrics

In our final report, we evaluated model performance using two key metrics: Mean Absolute Error (MAE) and R-squared (R^2). We chose MAE over RMSE because some CoCs had extremely high outlier counts of homeless individuals. RMSE would give those outlier errors much more weight than smaller, more typical errors, which could distort the evaluation of the model's overall performance. MAE treats all errors equally and gives a more balanced view.

- MAE (Mean Absolute Error): Indicates the average magnitude of prediction errors, showing how far off the model's predictions are from actual values

- R^2 (R-squared): Reflects how well the model explains the factors influencing increases or decreases in homelessness

Below is a matrix that illustrates how to interpret models based on their MAE and R^2 scores:

	Low MAE	High MAE
High R^2	Model **captures key drivers** of homelessness and makes **accurate predictions** Use to plan what factors to address to reduce homelessness & resource allocation quantities	Model **sees trends** but struggles with exact counts Use to see where to focus efforts but not how much to allocate
Low R^2	Predictions are **accurate,** but model **doesn't know why**	Model is both **inaccurate** and **uninformative**

Use for resource allocation, but not helpful for understanding what factors to direct strategy and policy towards	Do not use; unreliable for policy planning and resource allocation

Total Beds as a Feature

We chose to exclude total beds as an input feature in our models, even though doing so resulted in lower overall performance. We reasoned that total beds represent a response to homelessness, not a cause — they reflect existing demand rather than serving as a predictive factor. Including this feature risked introducing a feedback loop, where the model's predictions could become influenced by prior outputs, ultimately reinforcing existing trends rather than uncovering underlying drivers of homelessness. Additionally, we observed that total beds had the highest feature importance in our XGBoost model, raising concerns about the model becoming overly dependent on a single reactive variable rather than learning from more meaningful predictors.

Model #1: Why we Built a Linear Regression

We built a linear regression model as our initial approach because it is simple to implement, fast to train, and highly interpretable. These qualities make it an ideal starting point for modeling. Linear regression also serves as a strong baseline, as it assumes a straightforward, linear relationship between the input features and the target variable. By establishing this baseline, we can more effectively evaluate whether more complex models are truly providing additional predictive value beyond what a simple linear model can achieve.

Model #2: Why we Built a Neural Network

Homelessness is influenced by numerous factors (e.g., economic conditions, housing affordability, unemployment, average temperature). Neural networks excel at detecting intricate, nonlinear interactions between these variables.When working with large datasets containing multiple predictors, neural networks efficiently process and learn from high-dimensional inputs. Neural networks outperform simpler models because they can automatically adjust many parameters and weights in order to get the highest accuracy.

Fine Tuning the Neural Network

For the fine tuning of the neural networks model, we implemented several strategies. Regularization played a crucial role—we applied L2 weight decay with a rate of 0.01 to all hidden layers, which helped prevent overfitting by penalizing excessively large weights. This ensured the model would not rely too heavily on any single feature, promoting better generalization.

Dropout layers further enhanced robustness by randomly deactivating 20% of neurons during training. This technique forced the network to develop redundant trends, making it less dependent on specific neurons and more resilient to noise in the data. Alongside dropout, we employed early stopping with a patience of 20 epochs, halting training if validation performance plateaued. This not only saved computation time but also prevented the model from memorizing training data at the expense of real-world applicability.

We used a sequential structure with three hidden layers of decreasing width (64, 32, and 16 neurons), each activated by ReLU to capture nonlinear patterns efficiently. The output layer used a linear activation, appropriate for our regression task. Training was conducted with the Adam optimizer, known for its adaptive learning rates, and a moderate batch size of 32 to balance convergence speed and stability.

Before training, we standardized all features to ensure equal contribution—a critical step given neural networks' sensitivity to input scales. Together, these adjustments allowed the model to uncover complex relationships between socioeconomic factors and homelessness without succumbing to overfitting or instability.

Model #3: Why we Built an XGBoost model

We built an XGBoost model to leverage the strengths of gradient-boosted decision trees to capture complex, nonlinear patterns and interactions among our predictors without sacrificing training efficiency or overfitting control. XGBoost iteratively combines many "weak" tree learners, each one focusing on the residual errors of its predecessors, which allows it to flexibly model relationships that neither a simple linear model nor a standard neural network can easily capture.

Additionally, XGBoost offers built-in regularization (L1 and L2 penalties), robust handling of missing values, and automated feature importance scoring. These features help prevent overfitting, streamline preprocessing, and provide clear insights into which factors (e.g., unemployment rates, housing costs, temperature fluctuations) drive homelessness predictions. Finally, XGBoost's highly optimized implementation delivers fast training and scalable performance, making it an ideal choice when balancing predictive power and interpretability.

Finetuning XGBoost Model:

To maximize our XGBoost model's predictive power while guarding against overfitting, we partitioned the data into an 80 % training set and a 20 % hold-out test set. We then wrapped an **XGBoostRegressor**—configured for squared‑error regression—in a five-fold cross-validated grid search that optimized for the lowest mean squared error. Throughout 288 candidate configurations, we varied tree depth (**Max_Depth** of 2 or 3), number of boosting rounds (at 100, 200 or 500), and the learning rate (fixed at 0.01), split-gain threshold (**Gamma** of 0 or 1), and both L1 (**reg_Alpha** of 0, 0.1 or 1) and L2 (**reg_lambda** of 1 or 10) regularization terms.

After evaluating each hyperparameter combination on the validation folds, we selected the configuration that delivered the lowest average MSE. We retrained this "best" model on the whole training set. Then we assessed its performance on training and hold-out data—reporting MSE, RMSE, MAE, and R^2—to confirm that our regularization and subsampling choices struck the right balance between bias and variance. This thorough tuning process harnessed XGBoost's gradient-boosting mechanics, built-in penalties, and efficient tree-based learning to yield a robust predictor of sheltered homelessness.

Data Overview

Data Caveats

When merging the corrected PIT data with the ACS dataset, we lost approximately 150 CoC-year pairs, reducing the total number of usable rows to 6,386. We believe this data loss stems from differences in how CoCs are defined across the two sources. The PIT data reflects CoC boundaries that may shift from year to year, whereas our ACS dataset relies on fixed census tract-to-CoC mappings — specifically, 2019 tract definitions for years 2007–2019 and 2022 tracts for years 2020–2023. This mismatch likely caused some CoCs present in the PIT data to have no corresponding match in the ACS-derived dataset, leading to dropped entries during the merge process.

The following CoC-year pairs were dropped from the ACS dataset:

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[('AR-503', 2007), ('AR-503', 2008), ('AR-503', 2009), ('AR-503', 2010),
('AR-504', 2011), ('AR-504', 2012), ('AR-504', 2013), ('AR-504', 2014),
('AR-504', 2015), ('AR-504', 2016), ('AR-504', 2017), ('AR-504', 2018),
('AR-504', 2019), ('CA-527', 2007), ('CA-527', 2008), ('CA-527', 2009),
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('MA-517', 2019), ('MD-500', 2007), ('MD-500', 2008), ('MD-500', 2009),
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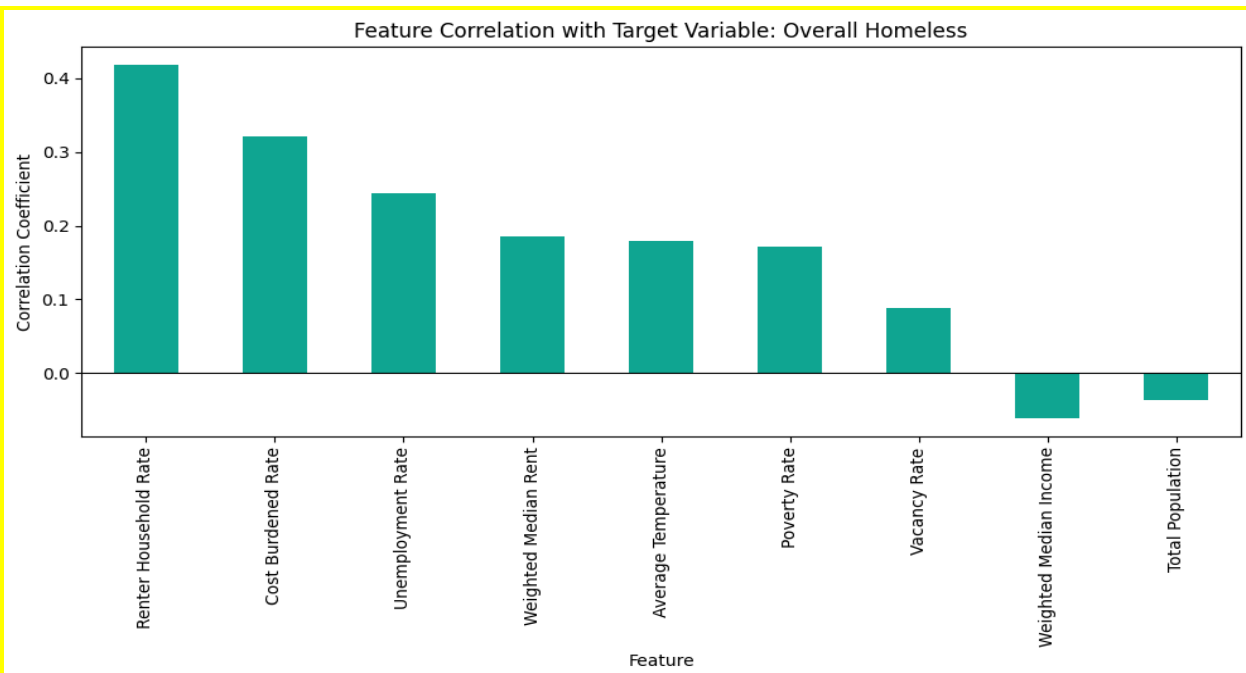
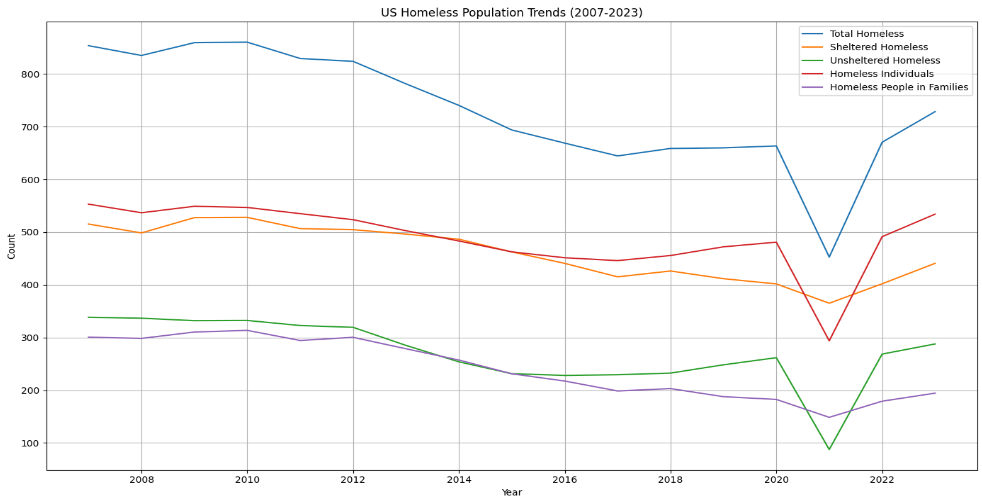
When we recalculated the population, rent, and income values using the corrected population-weighted averaging method, we found that our results differed slightly from the values reported in last semester's dataset—even when replicating their (incorrect) summing approach. While our updated rent and income values are now accurate, the numbers we obtained using the previous method still did not exactly match those from last semester. We initially suspected the difference might be due to last semester's team using 2022 ACS data as a proxy for 2023, whereas we had access to actual 2023 ACS data. However, this did not fully explain the discrepancy. Unfortunately, without access to the original data files or specific links used by the previous team, we were unable to trace the exact source of the mismatch.

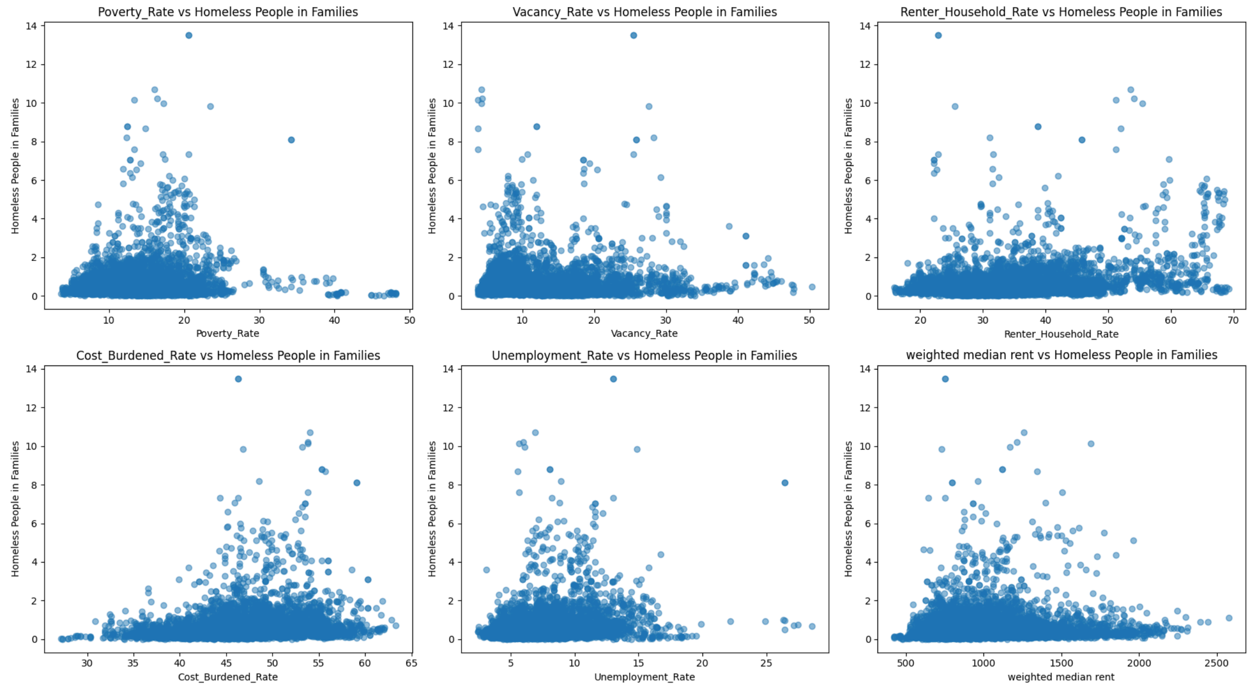
Exploratory Data Analysis



This plot shows how different factors are linked to homelessness. The Renter Household Rate has the strongest connection, with a correlation of about 0.4, meaning places with more renters tend to have more homelessness. Next, the Cost Burdened Rate and Unemployment Rate, at around 0.3 and 0.25, show that people struggling to pay bills or find jobs are more likely to end up homeless. Weighted Median Rent and Average Temperature are also tied to homelessness, with correlations around 0.2—higher rents and maybe warmer weather seem to play a role. Poverty Rate and Weighted Vacancy Rate have smaller connections, around 0.15 and 0.1. On the flip side, Weighted Median Income and Total Population have tiny negative correlations, around -0.05, so higher incomes and bigger populations might lower homelessness a bit. Overall, this chart makes it clear that things like housing costs and jobs matter a lot when it comes to homelessness.



This second plot shows how homelessness in the US changed from 2007 to 2023, breaking it down into different groups. The blue line, total homelessness, starts around 800,000 in 2007, drops steadily to about 600,000 by 2016, stays flat for a few years, dips sharply to around 500,000 in 2020—likely due to pandemic-related eviction moratoriums—and then spikes back up to over 700,000 by 2023. The red line, homeless individuals, follows a similar trend, starting at 500,000, falling to 400,000 by 2016, dipping to 350,000 in 2020, and jumping to 500,000 by 2023. The orange line, unsheltered homeless, stays steady around 400,000 until 2020, then drops to 300,000 before climbing back to 400,000. The green line, sheltered homeless, hovers around 200,000 for most of the period but drops to 150,000 in 2020 before rising again. Lastly, the purple line, homeless people in families, stays pretty stable around 200,000, with a slight dip in 2020. This graph shows that while homelessness overall went down for a while, it shot up again after 2020, with more people ending up unsheltered.



This third plot shows how different factors relate to the number of homeless people in families, and honestly, they tell an interesting story. Each dot represents a data point, with the x-axis showing things like poverty rate, vacancy rate, renter household rate, cost burdened rate, unemployment rate, and weighted median rent, while the y-axis is always the number of homeless people in families. Most of the dots are clustered near the lower end of the x-axis, meaning a lot of the data points have lower values for these factors, and the number of homeless family members is often below 6. But there are some outliers—dots spread out further along the x-axis, like in the poverty rate plot where some areas have a rate as high as 50%, or in the weighted median rent plot where rent goes up to 2500. These outliers often correspond to higher numbers of homeless family members, sometimes reaching up to 14. This suggests that in places where things like poverty, high rent, or unemployment are more extreme, the number of homeless people in families tends to be higher, though it's not a perfect pattern since most areas don't see such high numbers.

Results

Model #1: Linear Regression

<u>Results before normalizing target variables</u>

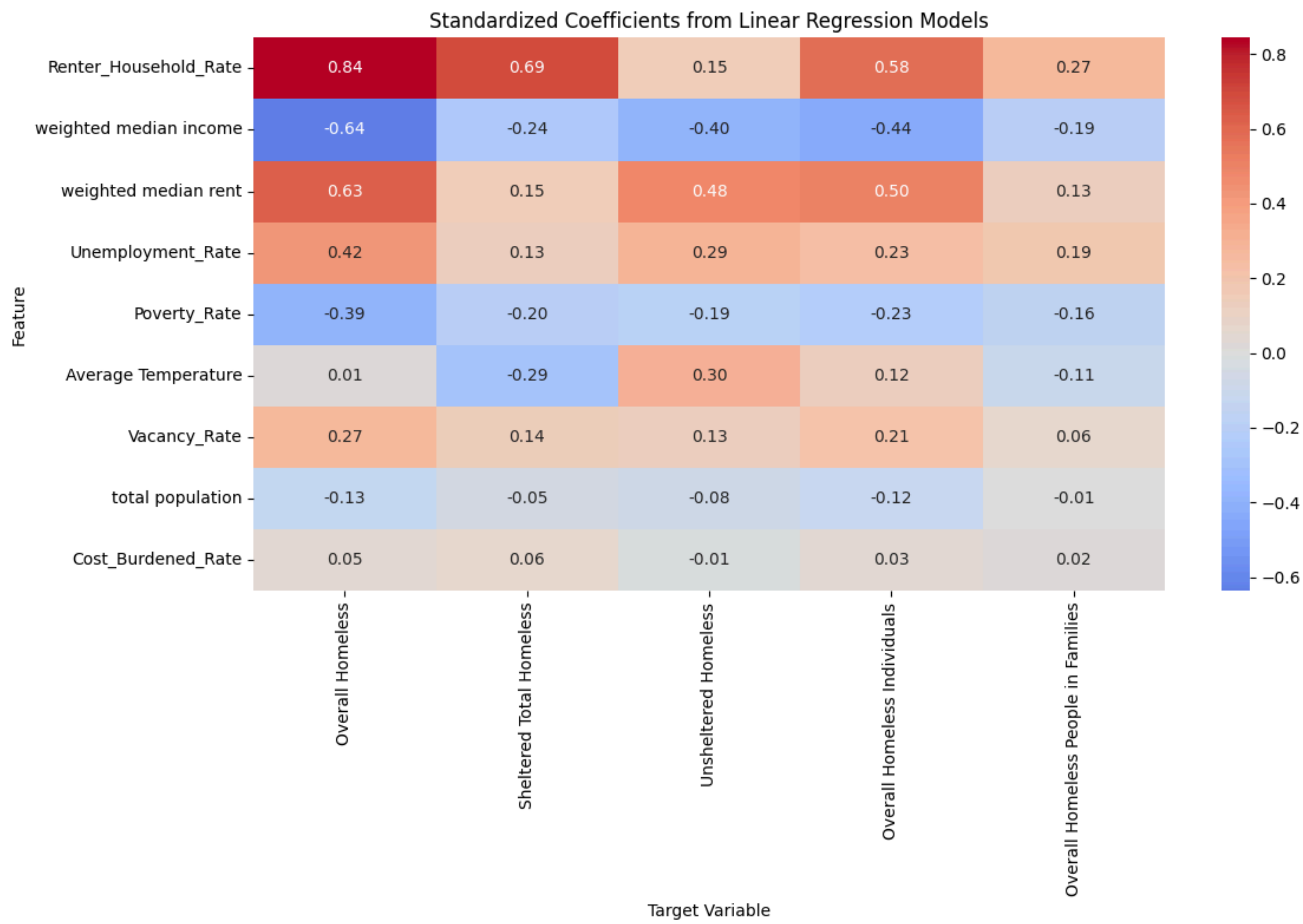
Target Variables	**R-Squared**	**MAE**

Overall Homeless Individuals	0.45	855.76
Overall Homeless	0.44	1363.82
Sheltered Homeless	0.33	967.79
Overall Homeless People in Families	0.28	611.55
Unsheltered Homeless	0.25	600.89

Results after normalizing target variables (per 1000 people)

Target Variables	**R-Squared**	**MAE**
Overall Homeless People in Families	0.1247	0.43
Overall Homeless Individuals	0.2641	0.81
Unsheltered Homeless	0.2025	0.74
Sheltered Homeless	0.2936	0.61
Overall Homeless	0.2635	1.12

The linear regression did poorly with and without the normalizing of target variables. To better understand the linear relationships between features and each target variable, we performed a full regression analysis using standardized linear regression models with normalized targets. The plot below displays the standardized coefficients from five separate linear models — one for each target. These coefficients reflect the direction (positive or negative) and relative strength of each feature's linear influence on the target. Standardizing the features ensures that the magnitude of the coefficients is comparable across variables. This visualization helps identify which features have the strongest linear associations with homelessness-related outcomes — and whether those relationships are positive or negative.



Model #2: Neural Network

<u>Results before normalizing target variables</u>

Target Variables	**R-Squared**	**MAE**
Overall Homeless	0.9396	488.55
Overall Homeless Individuals	0.8589	364.28
Overall Homeless People in Families	0.8898	358.3

Unsheltered Homeless	0.6105	350.69
Sheltered Homeless	0.99	167.9

Results after normalizing target variables (per 1000 people)

Target Variables	**R-Squared**	**MAE**
Overall Homeless People in Families	0.537	0.27
Overall Homeless Individuals	0.7156	0.52
Unsheltered Homeless	0.5713	0.51
Sheltered Homeless	0.8183	0.28
Overall Homeless	0.6976	0.69

Model #3: XGBoost

Results before normalizing target variables

Target Variables	**R-Squared**	**MAE**
Overall Homeless	0.91	604
Overall Homeless Individuals	0.828	440.8
Overall Homeless People in Families	0.944	237.6
Unsheltered Homeless	0.669	361.6
Sheltered Homeless	0.964	360

Results after normalizing target variables (per 1000 people)

Target Variables	**R-Squared**	**MAE**

Overall Homeless People in Families	0.3241	0.38
Overall Homeless Individuals	0.5929	0.64
Unsheltered Homeless	0.4915	0.56
Sheltered Homeless	0.5540	0.52
Overall Homeless	0.5257	0.92

Conclusion and Next Steps:

This semester, we successfully found and fixed underlying issues in the previous team's dataset to build models trained on the correct factors of homelessness across the nation. By using a linear regression model as our baseline, we learned more about the nature of the data and built two additional models that we thought would perform well with the complex, non-linear relationships in our dataset. Ultimately, we were very happy with the metrics of our final tuned models, with the XGBoost and Neural Network outperforming models from last semester, while having the more complex target variables fixed for per capita values. We recommend that the clients save the neural network models for model deployment, as they were the most accurate regarding R-squared and MAE. Finally, for the next steps of this project, we suggest going back into the ACS and PIT data to add back the CoC pairs that were lost during merging, and to look into additional datasets with features that can help better predict Homelessness rates across the nation.